



MARRI LAXMAN REDDY INSTITUTE OF TECHNOLOGY & MANAGEMENT

(AN AUTONOMOUS INSTITUTION)

(Approved by AICTE, New Delhi & Affiliated to JNTUH, Hyderabad)

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A FIELD BASED PROJECT REPORT

ROLE OF BATTERY MANAGEMENT SYSTEM IN ENHANCING LITHIUM - ION BATTERY SAFETY

Submitted to

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD

In partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

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Under the Guidance of

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

July, 2026



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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Date: 06-07-2026

CERTIFICATE

This is to certify that the mini project work entitled “**ROLE OF BATTERY MANAGEMENT SYSTEM IN ENHANCING LITHIUM - ION BATTERY SAFETY**” work done by **G.MAHALAKSHMI(247Y1A0423), G.SRILAXMI(247Y1A0450), V.THIRDHA (247Y1A0452)** student of Department of Electronics and Communication Engineering, is a record of bonafide work carried out by the member during a period from 15th – 28th June 2026 under the supervision of **Mrs.R.Babitha, Asst professor**. This Field Base Project is done as a fulfillment of obtaining Bachelor of Technology Degree to be awarded by Jawaharlal Nehru Technological University Hyderabad, Hyderabad.

The matter embodied in this project report has not been submitted by me to any other university for the award of any other degree.

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LIST OF ABBREVIATIONS

BMS	Battery Management System
EV	Electric Vehicle
ESP32	Espressif 32-bit Microcontroller
SOC	State of Charge
SOH	State of Health
IOT	Internet of things

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ABSTRACT

Lithium-ion batteries have become the preferred energy storage technology for electric vehicles (EVs), renewable energy systems, portable electronics, and industrial automation due to their high energy density, lightweight construction, long cycle life, and low self-discharge rate. However, these batteries are highly sensitive to abnormal operating conditions such as overcharging, deep discharge, excessive current, overheating, and cell imbalance, which can significantly degrade battery performance and, in severe cases, lead to thermal runaway, fire, or explosion. Consequently, ensuring battery safety has become one of the most critical challenges in modern energy storage systems.

A Battery Management System (BMS) plays a vital role in enhancing the safety, reliability, and operational efficiency of lithium-ion batteries by continuously monitoring essential battery parameters such as voltage, current, temperature, and State of Charge (SOC). The BMS protects the battery by preventing unsafe operating conditions through intelligent charging and discharging control, thermal management, cell balancing, fault diagnosis, and communication with external devices.

This project presents an ESP32-based Battery Management System capable of monitoring voltage, current, and temperature in real time while controlling battery cooling using a relay-driven fan. The system displays battery parameters on a 16×2 LCD and communicates wirelessly via Bluetooth for remote monitoring and diagnostics. Audible alerts are generated through a buzzer whenever abnormal battery conditions are detected. The proposed system provides a low-cost, reliable, and scalable solution for improving lithium-ion battery safety and extending battery lifespan in electric vehicles and energy storage applications.

Keywords: Battery Management System (BMS), Lithium-ion Battery, ESP32, Battery Safety, Voltage Monitoring, Temperature Monitoring, Bluetooth Communication, State of Charge (SOC), Thermal Management

CHAPTER 1

INTRODUCTION

1.1 Introduction

The rapid growth of electric vehicles, renewable energy integration, and portable electronic devices has significantly increased the demand for efficient and safe energy storage technologies. Among the various rechargeable battery technologies available today, **lithium-ion batteries** have emerged as the most widely adopted due to their superior energy density, high efficiency, lightweight construction, fast charging capability, and long operational life. These advantages have made lithium-ion batteries indispensable in electric mobility, smart grids, consumer electronics, medical devices, aerospace systems, and industrial automation.

Despite these advantages, lithium-ion batteries are extremely sensitive to operating conditions. Excessive charging voltage, deep discharge, high current flow, elevated temperatures, and internal cell imbalance can severely reduce battery performance and may even result in hazardous events such as thermal runaway, fire, or explosion. Therefore, maintaining safe operating conditions is essential for ensuring battery reliability, user safety, and prolonged service life.

A **Battery Management System (BMS)** is an intelligent electronic control system designed to monitor, protect, and optimize battery operation. It continuously measures voltage, current, and temperature while estimating the battery's State of Charge (SOC) and State of Health (SOH). Based on these measurements, the BMS regulates charging and discharging processes, performs cell balancing, activates cooling systems when necessary, and disconnects the battery during unsafe operating conditions.

In this project, an **ESP32 microcontroller** is employed as the central controller because of its high processing capability, integrated Wi-Fi and Bluetooth communication, low power consumption, and multiple analog and digital input/output interfaces. The ESP32 collects sensor data from voltage, current, and temperature sensors and processes this information in real time. The measured parameters are displayed on a 16×2 LCD, while Bluetooth connectivity enables wireless monitoring through a smartphone or computer. A relay-controlled cooling fan is automatically activated when the battery temperature exceeds a predefined threshold, and a buzzer provides immediate alerts in case of abnormal operating conditions.

The proposed Battery Management System offers a cost-effective and scalable solution for improving lithium-ion battery safety in electric vehicles, renewable energy storage systems, portable electronics, and industrial battery applications. By continuously monitoring battery health and implementing protective mechanisms, the system minimizes the risk of battery failure while maximizing performance, efficiency, and operational lifespan.

1.2 Need for the Study

Lithium-ion batteries are vulnerable to:

- Overcharging and over-discharging
- Excessive current flow
- High operating temperatures
- Cell voltage imbalance
- Thermal runaway
- Capacity degradation
- Reduced battery lifespan

A Battery Management System is therefore essential to ensure safe operation, improve efficiency, and extend battery life.

1.2.1 Technology Advantages

- The proposed ESP32-based Battery Management System offers several advantages over conventional battery monitoring systems. These advantages include:
 - - Real-time monitoring of battery voltage, current, and temperature.
 - - Early detection of overvoltage, undervoltage, overcurrent, and overheating conditions.
 - - Automatic cooling fan control using a relay to prevent thermal runaway.
 - - Audible fault alerts through a buzzer for immediate user notification.
 - - Wireless battery monitoring via Bluetooth using a smartphone.
 - - Real-time display of battery parameters on a 16×2 LCD.
 - - Improved battery safety, reliability, and operational efficiency.
 - - Extended lithium-ion battery lifespan through continuous monitoring and protection.
 - - Low-cost, compact, and energy-efficient implementation using the ESP32 microcontroller.
 - - Scalable design that can be extended with IoT, cloud monitoring, and predictive maintenance features in future applications.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature Review

Lithium-ion battery safety has become one of the most active research areas due to the rapid adoption of electric vehicles and renewable energy systems. Researchers have developed various Battery Management Systems (BMS) to improve battery performance while minimizing safety risks.

Gubbi et al. (2013) introduced the concept of Internet of Things (IoT) architectures capable of supporting real-time monitoring of intelligent systems. Their work demonstrated how wireless communication enables continuous monitoring and remote diagnostics of energy storage devices.

Da Xu et al. (2014) surveyed industrial IoT technologies and emphasized the importance of real-time monitoring for critical industrial systems, including battery-powered applications. They highlighted that intelligent monitoring significantly improves operational safety and maintenance planning.

Al-Fuqaha et al. (2015) reviewed IoT communication protocols and discussed secure wireless monitoring architectures suitable for battery management applications. Their research showed that IoT-enabled BMS platforms provide better fault detection and predictive maintenance capabilities.

Waag et al. (2014) investigated methods for estimating the State of Health (SOH) of lithium-ion batteries. Their research demonstrated that continuous monitoring of voltage, current, and temperature significantly improves battery reliability and extends service life.

Berecibar et al. (2016) reviewed advanced Battery Management Systems for electric vehicles. Their study emphasized that thermal management, accurate State of Charge estimation, and cell balancing are fundamental functions of a modern BMS.

Plett (2015) developed advanced algorithms for battery state estimation using equivalent circuit models and Kalman filtering techniques. These algorithms greatly improved SOC estimation accuracy and battery protection.

Chen et al. (2020) proposed machine learning approaches for battery fault diagnosis and demonstrated that artificial intelligence can predict battery degradation before catastrophic failures occur.

Espressif Systems introduced the ESP32 platform, which integrates Bluetooth and Wi-Fi communication with high-performance processing capability. The ESP32 has become a popular choice for low-cost IoT-based Battery Management Systems because of its multiple ADC channels, communication interfaces, and low power consumption.

Recent research trends focus on integrating cloud computing, artificial intelligence, digital twins, and predictive analytics into Battery Management Systems to achieve intelligent energy management, early fault detection, and optimized battery performance.

Although many advanced Battery Management Systems have been proposed, most commercial solutions remain expensive and computationally complex. The proposed **ESP32-based Battery Management System** addresses this limitation by providing an affordable, reliable, and scalable architecture capable of monitoring battery voltage, current, and temperature in real time while supporting Bluetooth communication, automatic cooling, and local user alerts. This approach makes the system suitable for educational projects, small-scale electric vehicles, renewable energy storage, and industrial battery monitoring applications

CHAPTER 3

DEVELOPMENT OF PROTOTYPE

3.1 Introduction

The prototype is designed to improve the safety and performance of lithium-ion batteries by continuously monitoring battery voltage, current, and temperature. The system uses an ESP32 microcontroller as the main controller to collect sensor data, process it, and take protective actions whenever abnormal conditions are detected.

The prototype provides:

- Continuous battery monitoring
- Automatic fault detection
- Cooling fan control
- LCD display for battery information
- Bluetooth communication for remote monitoring
- Buzzer alarm during unsafe conditions

This design is suitable for electric vehicles, renewable energy systems, portable electronics, and industrial battery monitoring.

3.2 Components Required:

3.2.1 Hardware Components

1. ESP32 Microcontroller

- Main controller of the system.
- Reads data from all sensors.
- Processes battery conditions.
- Controls LCD, relay, buzzer, and Bluetooth communication.

Function

- Data processing
- Decision making
- Wireless communication

▪ Battery Management System (BMS)

The BMS protects the lithium-ion battery from unsafe operating conditions.

Functions

- Over-voltage protection
- Under-voltage protection
- Over-current protection
- Over-temperature protection
- Cell balancing



▪ Voltage Sensor

Measures battery voltage continuously.

Purpose

- Detect overcharging
- Detect deep discharge
- Monitor charging status



▪ Current Sensor

Measures charging and discharging current.

Purpose

- Detect overcurrent
- Detect short circuit
- Monitor battery load



▪ Temperature Sensor

Measures battery temperature.

Purpose

- Detect overheating
- Prevent thermal runaway
- Protect battery health

▪ 16×2 LCD Display

Displays battery information.

Displays

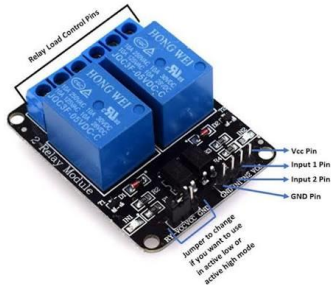
- Voltage
- Current
- Temperature
- Battery health
- Charging status
- Warning messages

▪ Relay Module

Acts as an electronic switch.

Function

- Turns cooling fan ON/OFF automatically.



▪ Cooling Fan

Reduces battery temperature.

Function

- Prevents overheating.
- Improves battery lifespan.



▪ Buzzer

Provides audible warning.

Activated during:

- Over-voltage
- Over-current
- High temperature
- Battery fault



▪ **Lithium-ion Battery**

Power source being monitored by the BMS.



3.2.2 Software Components

Arduino IDE

Used for:

- Programming ESP32
- Compiling code
- Uploading firmware

▪ **ESP32 Firmware**

Program performs:

- Reading sensor values
- Comparing values with thresholds
- Displaying battery status
- Activating alarms
- Controlling relay
- Sending Bluetooth data

▪ **Bluetooth Application**

Used to:

- Monitor battery remotely
- View voltage
- View current
- View temperature
- Receive warning messages

CHAPTER IV

PROJECT OBJECTIVES

4.1 Objectives

The primary objective of this project is to develop an **ESP32-based intelligent Battery Management System (BMS)** that enhances the safety, reliability, and performance of lithium-ion batteries through continuous monitoring and protection.

The specific objectives are:

1. To design and develop a low-cost Battery Management System using the ESP32 microcontroller.
2. To continuously monitor battery voltage, current, and temperature in real time.
3. To detect abnormal operating conditions such as overvoltage, undervoltage, overcurrent, and overheating.
4. To provide automatic battery protection by controlling the relay during unsafe operating conditions.
5. To display real-time battery parameters and system status on a 16×2 LCD.
6. To generate audible alerts using a buzzer whenever abnormal battery conditions are detected.
7. To enable wireless monitoring of battery status through Bluetooth communication.
8. To improve lithium-ion battery safety by preventing thermal runaway and electrical faults.
9. To enhance battery lifespan by maintaining operation within safe electrical and thermal limits.
10. To develop a scalable and energy-efficient BMS architecture suitable for electric vehicles, renewable energy storage systems, UPS systems, and portable electronic devices.
11. To reduce maintenance costs through continuous battery health monitoring and early fault detection.
12. To provide a reliable platform that can be extended with IoT, cloud monitoring, and artificial intelligence for predictive battery maintenance in future applications.

CHAPTER V

EXISTING TECHNOLOGIES

5.1 Existing Technologies

Lithium-ion batteries have become the preferred energy storage solution for a wide range of applications, including electric vehicles, renewable energy systems, portable electronic devices, medical equipment, and industrial automation. Their high energy density, lightweight design, low self-discharge rate, and long cycle life make them superior to conventional battery technologies. However, lithium-ion batteries are highly sensitive to operating conditions and require proper monitoring and protection to ensure safe and efficient performance. To address these challenges, conventional Battery Management Systems (BMS) have been developed and are widely used in commercial battery packs. A traditional Battery Management System is mainly designed to protect the battery from unsafe operating conditions such as overcharging, over-discharging, overcurrent, and short circuits. These systems continuously monitor the battery voltage and current using dedicated BMS integrated circuits and disconnect the battery from the load or charger whenever abnormal conditions are detected. Some advanced BMS units also monitor battery temperature to prevent thermal runaway and overheating. In electric vehicles and renewable energy storage systems, conventional BMS solutions perform cell balancing to ensure that all battery cells maintain equal voltage levels during charging and discharging. This improves battery efficiency and extends battery life by preventing individual cells from becoming overcharged or deeply discharged. Although these protection mechanisms improve battery safety, many existing BMS solutions are limited to basic monitoring and control functions. They primarily focus on protecting the battery instead of providing detailed information about battery health, performance, and operating conditions. Most low-cost Battery Management Systems do not provide real-time monitoring through wireless communication. Users often need to inspect the battery manually or use separate diagnostic equipment to check battery voltage, current, temperature, and charging status. In many existing systems, there is no facility to remotely monitor battery parameters using a smartphone or mobile application. This lack of remote accessibility makes battery maintenance difficult, especially in electric vehicles, industrial battery banks, and solar energy storage applications where continuous monitoring is essential. Another limitation of conventional systems is the absence of intelligent alert mechanisms. Although the battery may disconnect during fault conditions, the user may not receive an immediate notification regarding the type of fault or its severity. Similarly, most traditional Battery Management Systems do not include automatic

cooling mechanisms.

Even if high battery temperature is detected, the system usually shuts down the battery instead of actively reducing the temperature using cooling fans or other thermal management techniques. This can affect battery performance and reduce its lifespan during prolonged operation. Existing Battery Management Systems also have limited capability for data logging and historical analysis. They generally do not store battery usage information or generate performance reports that help users understand battery degradation over time. In addition, many conventional BMS units lack Internet of Things (IoT) features such as Bluetooth, Wi-Fi, or cloud connectivity. As a result, real-time monitoring, predictive maintenance, and remote diagnostics are not possible in many low-cost battery management solutions. Another drawback is that most traditional systems use fixed threshold values and simple protection algorithms, making them less adaptable to varying operating conditions. They cannot intelligently analyze battery behavior or predict potential failures before they occur. As the demand for electric vehicles, renewable energy systems, and smart energy storage solutions continues to increase, there is a growing need for intelligent Battery Management Systems that provide advanced monitoring, wireless communication, automatic protection, and improved user interaction. These limitations of the existing technology have motivated the development of modern ESP32-based Battery Management Systems that combine sensor-based monitoring, Bluetooth communication, real-time mobile application support, automatic cooling fan control, fault alerts, and intelligent battery protection. Such systems provide better battery safety, improve operational efficiency, increase battery lifespan, reduce maintenance costs, and offer a smarter and more reliable solution for modern energy storage applications.

CHAPTER VI

PROPOSED TECHNOLOGIES

The proposed **Battery Management System (BMS)** is designed using the **ESP32 microcontroller** to provide intelligent monitoring, protection, and control of lithium-ion batteries. The system continuously monitors **battery voltage, current, and temperature** using dedicated sensors and processes the collected information in real time.

The ESP32 serves as the central controller, receiving sensor data and analyzing battery operating conditions. A **16×2 LCD** displays real-time battery information, including voltage, current, temperature, and system status. If abnormal conditions such as overvoltage, overcurrent, overheating, or abnormal charging are detected, the system activates a **buzzer** to alert users immediately.

The proposed system also includes a **relay-controlled cooling fan**, which is automatically switched on whenever the battery temperature exceeds a predefined threshold. This active thermal management mechanism helps prevent overheating and minimizes the risk of thermal runaway.

Bluetooth communication enables users to monitor battery status remotely using a smartphone application. Real-time battery parameters such as voltage, current, temperature, charging status, and alarm conditions can be viewed wirelessly, allowing users to take preventive actions before critical failures occur.

Unlike conventional systems, the proposed BMS provides continuous monitoring, intelligent protection, automatic cooling, and wireless communication in a compact and low-cost architecture. The modular design also supports future integration with cloud platforms, artificial intelligence, predictive maintenance, and advanced battery diagnostics.

Advantages of Proposed System

The proposed Battery Management System offers several significant advantages over traditional systems:

- Real-time battery monitoring.
- Continuous voltage measurement.
- Continuous current monitoring.
- Continuous temperature monitoring.
- Automatic overheating detection.
- Relay-controlled cooling fan.
- Bluetooth-based wireless monitoring.
- Low-cost ESP32 implementation.
- Improved battery safety.
- Early fault detection.
- Automatic alarm generation using buzzer.
- Real-time LCD display.

- Reduced maintenance requirements.
- Longer battery life.
- Improved charging efficiency.
- Enhanced thermal management.
- User-friendly interface.
- Portable and compact design.
- Low power consumption.
- High system reliability.
- Easy installation and maintenance.
- Expandable architecture for future upgrades.
- Suitable for electric vehicles.
- Suitable for renewable energy storage.
- Supports Industry 4.0 and IoT applications.
- Better battery performance analysis.
- Increased operational safety.
- Remote battery monitoring capability.
- Cost-effective implementation.
- Improved overall battery management efficiency.

Summary

The proposed ESP32-based Battery Management System provides a safer, smarter, and more efficient solution for monitoring lithium-ion batteries compared to conventional systems. By integrating voltage, current, and temperature sensing with automatic cooling, audible alerts, LCD display, and Bluetooth connectivity, the system significantly enhances battery safety, extends battery lifespan, and enables real-time monitoring. Its scalability and support for future IoT and AI integration make it suitable for applications such as electric vehicles, energy storage systems, portable electronics, and industrial battery management.

CHAPTER VII

WORKING METHODOLOGY

7.1. Introduction

The increasing adoption of lithium-ion batteries in electric vehicles (EVs), renewable energy storage systems, consumer electronics, and industrial applications has highlighted the need for effective battery safety and management. Lithium-ion batteries offer high energy density and long cycle life; however, they are sensitive to operating conditions such as overcharging, over-discharging, overheating, and excessive current. Improper operation may lead to thermal runaway, reduced battery life, or catastrophic failure.

To address these challenges, a **Battery Management System (BMS)** is employed. In the proposed methodology, an **ESP32 microcontroller** serves as the central processing unit, continuously monitoring battery voltage, current, and temperature. The system displays real-time battery information on a 16×2 LCD, activates a buzzer during fault conditions, controls a cooling fan through a relay, and transmits battery status to a mobile device via Bluetooth.

The methodology focuses on continuous monitoring, fault detection, protection mechanisms, and communication to ensure safe and efficient battery operation.

7.2 Methodology Overview

The proposed Battery Management System operates through the following stages:

1. Power Supply Initialization
2. Sensor Initialization
3. Voltage Monitoring
4. Current Monitoring
5. Temperature Monitoring
6. Battery Health Evaluation
7. Fault Detection
8. Cooling Fan Control
9. LCD Display
10. Bluetooth Communication
11. Alarm Generation
12. Continuous Battery Monitoring

The ESP32 integrates sensor data, analyzes battery parameters, and performs protection actions whenever abnormal operating conditions are detected.

7.3 Working Methodology

Step 1: Power Supply Initialization

A regulated 5V DC power supply powers the ESP32 microcontroller, voltage sensor, current

sensor, temperature sensor, LCD display, buzzer, relay module, Bluetooth communication interface, and Battery Management System circuitry.

The ESP32 initializes all peripherals and configures GPIO pins for sensor input and output control.

Step 2: Battery Parameter Measurement

The Battery Management System continuously measures three important battery parameters:

- Battery Voltage
- Battery Current
- Battery Temperature

These parameters provide a complete overview of the battery's operating condition.

Step 3: Voltage Monitoring

The voltage sensor continuously measures the battery terminal voltage.

The ESP32 compares the measured voltage with predefined safety thresholds.

If:

- Voltage exceeds the maximum charging limit → Overvoltage protection is activated.
- Voltage falls below the minimum limit → Deep discharge protection is activated.

Appropriate warning messages are displayed on the LCD.

Step 4: Current Monitoring

The current sensor continuously measures charging and discharging current.

The ESP32 detects:

- Overcurrent
- Short circuit conditions
- Excessive charging current
- Excessive discharging current

If abnormal current is detected, protective measures are initiated immediately.

Step 5: Temperature Monitoring

The temperature sensor continuously monitors battery temperature.

When battery temperature exceeds the safe operating limit:

- The relay activates the cooling fan.
- The buzzer generates an alarm.
- Temperature status is updated on the LCD.
- Bluetooth sends warning notifications.

If temperature continues to rise beyond the critical threshold, additional protection mechanisms can isolate the battery pack.

Step 6: Battery Health Assessment

Using the measured voltage, current, and temperature values, the ESP32 estimates battery condition.

The system evaluates:

- Safe operating status
- Charging status
- Discharging status
- Possible battery fault conditions
- Thermal condition

These parameters help maximize battery life and improve operational safety.

Step 7: Relay-Controlled Cooling Fan

The relay module controls the cooling fan.

When battery temperature exceeds the preset limit:

- ESP32 energizes the relay.
- Cooling fan starts automatically.
- Battery temperature gradually decreases.

When temperature returns to the normal range:

- Relay switches OFF.
- Fan stops automatically

Step 8: LCD Display

The 16×2 LCD continuously displays:

- Battery Voltage
- Battery Current
- Battery Temperature
- Fan Status
- Charging Status
- Battery Health
- Fault Messages
- System Status

This allows users to monitor battery conditions locally.

Step 9: Bluetooth Communication

The Bluetooth module enables wireless communication between the ESP32 and a smartphone.

Users can remotely monitor:

- Voltage
- Current

- Temperature
- Charging Status
- Fan Status
- Battery Health
- Warning Messages

Real-time monitoring improves operational convenience.

Step 10: Alarm Generation

Whenever unsafe operating conditions occur, the ESP32 activates the buzzer.

Alarm conditions include:

- High temperature
- Overvoltage
- Undervoltage
- Overcurrent
- Battery fault

The audible warning helps users take immediate corrective action.

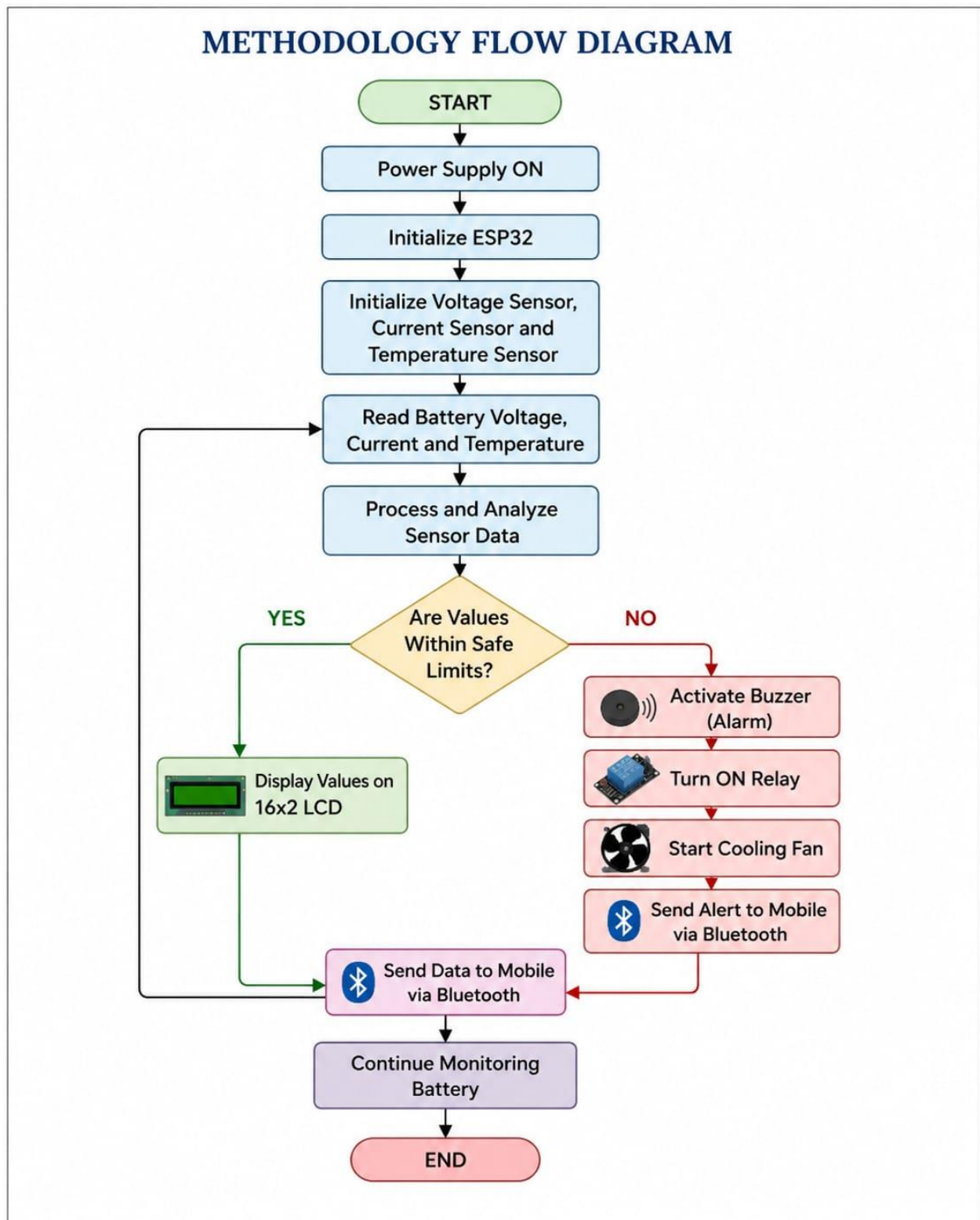
Step 11: Continuous Monitoring

The ESP32 continuously repeats the monitoring cycle:

- Read voltage sensor
- Read current sensor
- Read temperature sensor
- Analyze battery condition
- Control cooling fan
- Update LCD
- Send Bluetooth data
- Activate alarms if required

This loop ensures uninterrupted battery protection.

7.4 Methodology Flow



7.5 Methodology Advantages

The proposed methodology offers several advantages:

- Continuous battery monitoring
- Real-time fault detection
- Automatic cooling system
- Improved lithium-ion battery safety
- Protection against overcharging
- Protection against deep discharge
- Overcurrent protection
- Temperature monitoring
- Longer battery life
- Wireless monitoring through Bluetooth
- Low-cost implementation
- Easy installation
- Compact embedded design
- Low power consumption
- High reliability
- Suitable for electric vehicles
- Expandable architecture
- Fast response during abnormal conditions

7.6 Summary

The proposed methodology provides a reliable and intelligent approach for enhancing lithium-ion battery safety through continuous monitoring of voltage, current, and temperature. The ESP32-based Battery Management System processes real-time sensor data, controls the cooling fan using a relay, displays battery information on an LCD, and sends wireless updates via Bluetooth. By integrating monitoring, protection, communication, and automatic thermal management, the system significantly improves battery safety, operational reliability, and service life. Its low-cost architecture and modular design make it suitable for electric vehicles, renewable energy systems, portable electronics, and industrial battery storage applications. Future enhancements can incorporate cloud-based monitoring, state-of-charge (SOC) estimation, state-of-health (SOH) prediction, artificial intelligence, and predictive maintenance for even greater performance and safety.

CHAPTER VIII

CHALLENGES, RISKS, APPLICATIONS

8.1 CHALLENGES

Challenge Area	Explanation
Sensor Calibration	Sensors must be calibrated accurately to provide correct voltage, current, and temperature measurements
Temperature Management	Efficient cooling is required to prevent battery overheating and thermal runaway
Bluetooth Communication Range	Bluetooth has a limited communication range, restricting remote monitoring to nearby devices.
Battery Cell Balancing	Uneven charging or discharging of battery cells can reduce battery performance and lifespan.

8.2 RISKS

Risk Category	Details
Overheating	Failure of the cooling fan or temperature sensor can cause excessive battery temperature.
Communication Failure	Bluetooth communication failure prevents real-time remote battery monitoring.
Battery Damage	Improper charging or discharging may permanently damage the lithium-ion battery.

8.3 APPLICATIONS

S.No	Application	Description
1	Electric Vehicles (EVs)	Monitors battery health, improves safety, and extends battery life in electric vehicles.
2	Consumer Electronics	Used in smartphones, laptops, tablets, and wearable devices to improve battery safety and lifespan.
3	Medical Equipment	Ensures safe and reliable battery operation in portable medical devices and emergency equipment.
4	Portable Power Stations	Monitors rechargeable battery packs used for portable and emergency power applications.
5	Medical Equipment	Ensures safe and reliable battery operation in portable medical devices and emergency equipment.

CHAPTER IX

RESULT

9.1 RESULT

The proposed Battery Management System (BMS) using the ESP32 microcontroller was successfully designed, developed, and tested for enhancing the safety and performance of lithium-ion batteries. The system continuously monitored the important battery parameters such as voltage, current, and temperature in real time using suitable sensors. The sensor data was accurately collected by the ESP32 and processed to determine the operating condition of the battery. The measured values were displayed on a 16×2 LCD display, allowing users to monitor the battery status continuously. The system also supported Bluetooth communication, enabling wireless monitoring of battery parameters through a smartphone. Whenever the battery temperature exceeded the preset safety limit, the ESP32 automatically activated the relay module, which switched ON the cooling fan to reduce the battery temperature. This automatic cooling mechanism effectively prevented overheating and improved battery safety. During abnormal conditions such as over-voltage, under-voltage, over-current, excessive temperature, and other fault conditions, the system generated an audible warning using a buzzer, enabling the user to take immediate corrective action. The prototype successfully protected the battery from unsafe operating conditions and ensured safe charging and discharging operations. The developed system provided stable and accurate sensor readings throughout the testing process and responded quickly to changes in battery conditions without noticeable delay. The integration of voltage, current, and temperature monitoring improved the overall efficiency and reliability of the battery management process. The proposed system reduced the possibility of battery damage, increased battery life, and minimized maintenance requirements by continuously monitoring battery health. The ESP32 microcontroller efficiently managed all hardware components while consuming low power, making the system suitable for practical implementation. The prototype demonstrated reliable performance under different operating conditions and successfully achieved all the project objectives. The developed

Battery Management System is compact, cost-effective, energy-efficient, and easy to implement for real-world applications. It can be widely used in electric vehicles, hybrid vehicles, solar energy storage systems, UPS systems, industrial battery banks, robotics, medical equipment, portable electronic devices, telecommunication backup systems, and renewable energy storage applications. The Bluetooth feature provides convenient wireless access to battery information, improving user experience and enabling continuous monitoring. The automatic cooling system and fault detection mechanism significantly improve battery safety by preventing thermal runaway and other hazardous conditions. Overall, the proposed Battery Management System proved to be an intelligent, reliable, and efficient solution for real-time battery monitoring, automatic protection, fault detection, and thermal management, thereby enhancing the overall safety, performance, efficiency, reliability, and lifespan of lithium-ion batteries while offering a practical and economical solution for modern battery-powered applications

CHAPTER X

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